

The Results section: part 3

Fish 290

Tuesday February 11th

Posters are coming up!

- Poster session is Thursday Feb 20th (one week from Thursday).
- Attendance in class will be mandatory.
- Your poster is worth 20% of your final grade
- We will work on posters Thursday

How can you prepare now?

- Continue re-reading your chosen paper and get to know it well
- Observe the posters from actual studies (see Canvas) and past student posters. What works, what doesn't?

Important dates

- **Friday Feb 14:** Send me all your paper progress so far, including Intro/Methods/Results/figures+tables
- **Thursday Feb 20:** Poster presentations, posters due
- **Friday Feb 28:** Complete first draft of papers due (no bullet points, all sections completed**)
- **Thursday March 13th:** oral presentations
- **Thursday Mach 13th:** final papers due

** all sections = Title, Abstract, Introduction, Methods, Results, Discussion, Acknowledgements, References, any tables and figures

Making effective tables

Tables are appropriate when the precision of a value matters (like a p-value) and you have many variables to display.

Tables are also good for background information (comparing field sites, results from prior literature, or species)

For example:

- A list of species used in study and their locations and diets
- A list of relevant past studies and their key findings
- The results of multiple statistical tests

Making effective tables

We read from left-to-right, so put context-setting information on the left and dependent information on the right.

These tables show the same data. Which is easier to read?

Time (minutes)	Temperature (°C)
0	25
3	27
6	29
9	31
12	32
15	32

Temperature (°C)	Time (minutes)
25	0
27	3
29	6
31	9
32	12
32	15

Making effective tables

Be judicious about significant figures: just because Excel says the mean is 0.4684504, doesn't mean we need that many numbers.

Pay attention to how others report similar types of values.

These tables show the same results. Which is easier to read?

Mean	P-value
45.754	0.131543
31.157	0.032698
131.845	0.146644
14.064	0.0022465
113.254	0.1202454
9.015	0.791545

Mean	P-value
45.7	0.131
31.1	0.032
131.8	0.146
14.0	0.002
113.2	0.120
9.01	0.791

Making effective tables

Some more advice:

- Bold significant p-values so they stand out from non-significant values
- Similar values should match in the # of significant figures

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Mean	P-value
45.7	0.131
31.1	0.032
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113.2	0.120
9.0	0.791

Making effective tables

It is convention to use horizontal borders on the header row and last row, but not all borders.

These tables show the same data. Which is more pleasing to the eye?

Order	Variable A	Variable B	Variable C
Acipenseriformes	27	26	26
Amiiformes	1	1	0
Lepisosteiformes	7	7	5
Albuliformes	11	8	8
Anguilliformes	933	197	191
Elopiformes	9	6	6
Notacanthiformes	27	8	2
Lepidogalaxiiformes	1	1	0
Ateleopodiformes	13	6	6

Order	Variable A	Variable B	Variable C
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Notacanthiformes	27	8	2
Lepidogalaxiiformes	1	1	0
Ateleopodiformes	13	6	6

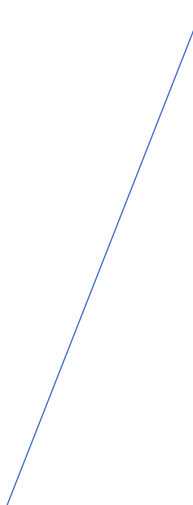


Your tables should look like
this

Order	Variable A	Variable B	Variable C
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Notacanthiformes	27	8	2
Lepidogalaxiiformes	1	1	0
Ateleopodiformes	13	6	6

Every label should be self explanatory, or explained in the caption.

Don't make the reader hunt the main text to interpret



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Notacanthiformes	27	8	2
Lepidogalaxiiformes	1	1	0
Ateleopodiformes **	13	6	6

* Excludes moray eels

** Three species had no data



Symbols are sometimes explained at the end of the table itself, other times in the table caption.

The importance of the caption

Each figure *and* table must have a fully self-explanatory caption (sometimes called a legend). The reader should be able to look at the figure or table and the caption, and understand what is being presented.

By convention, **figure captions** are usually placed **below** the image, whereas **table captions** are usually placed **above** the table.

Remember, the reader is busy. The abstract and figures/tables may be the only thing they read. Do not make the reader go through the entire paper to understand what is going on. Your captions are important to give the reader context!

What makes an effective caption?

Key points:

- Describes independent and dependent variables on a figure (x and y axes)
- Defines symbols or abbreviations used in figure.
- Specifies temporal bounds of the data (if necessary)
- **Briefly** indicates how data were collected (if necessary)
- Is fully self-explanatory
- Specifies if a pattern shown is stat. significant (if relevant)
- Just the facts (resist urge to interpret)

Things to avoid:

- Vague statements (“graph of fish population”)
- Repeating information already in figure (like a legend)
- Too short or too long.
- Interpretation (can be intrusive to the reader- they’re trying to see for themselves and don’t need you butting in!)

Making effective captions

Some examples of real captions, figures and tables.

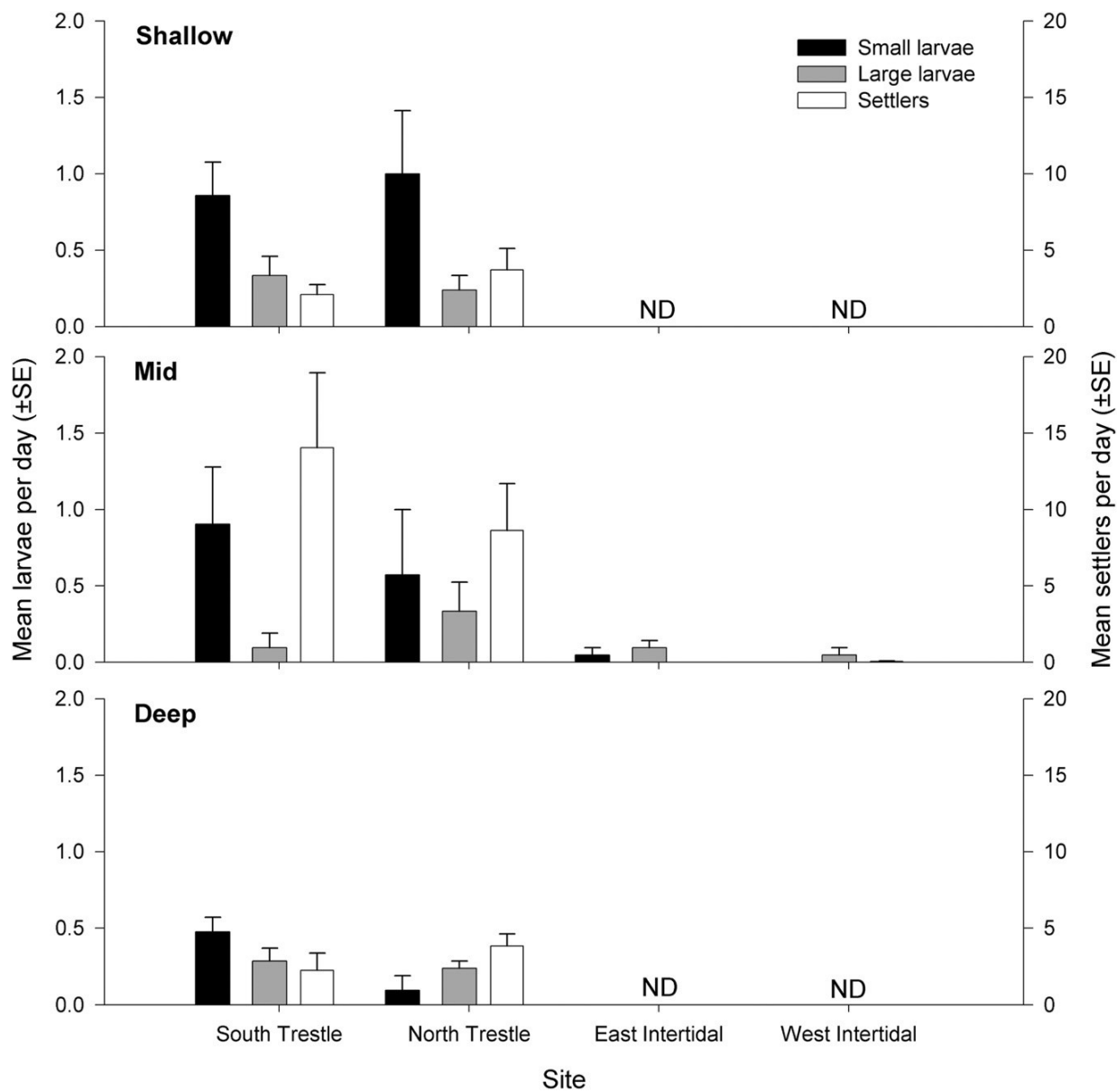


Figure 4. Spatial patterns of planktonic larval capture and settlement (mean $\pm$ 1 SE) at intertidal sites at the three depths (shallow: 0.2 m, middle: -0.3 m, and deep: -0.8 m). Larval capture data from passive traps are presented for small and large larvae for the week ending July 25, 2013. Settlement data are station means across the entire sampling interval June 13, 2013–August 1, 2013. ND, no data.

The effects of adult length and arrival date on individual reproductive success in wild steelhead trout (*Oncorhynchus mykiss*)

Todd R. Seamons, Paul Bentzen, and Thomas P. Quinn

Table 2. Number of adult male and female steelhead (N), their size range (SR, mm) and average length (L_{Avg} , mm), and their median arrival date (AD_M).

Brood year	Adults											
	Males				Females				Juveniles			
	N	SR	L_{Avg}	AD_M	N	SR	L_{Avg}	AD_M	N_{0+}	N_{1+}	N_{2+}	N_{NP}
1997	11	420–750	612.7	20 Mar. ^a	20	595–830	692.3	20 Mar. ^a	130	60	1	66
1998	35	600–735	649.6	20 Mar.	18	565–795	638.6	21 Mar.	117	22	2	23
1999	25	400–815	655.6	23 Mar.	27	600–740	658.3	23 Mar.	161	49		134
2000	64	430–785	645.4	23 Mar.	75	565–860	640.0	25 Mar.	245			13
Total	135				140				653	131	3	236

Note: For each brood year, the columns on the right side list the number of juveniles with at least one identified parent in each of three age classes: young-of-the-year (N_{0+}), yearling (N_{1+}), and 2-year-olds (N_{2+}). The rightmost column gives the number of juveniles from each brood year for which no parents were identified (N_{NP}).

^aEstimated.

Loss of native rocky reef biodiversity in Australian metropolitan embayments

Table 1
Pearson correlations between urban impact variables and exposure in the three cities analysed. SO = sewage outfall. NA = not assessed. Correlations >0.7 are in bold.

Location	Variable	Exposure	Population index	Invasive species	Distance from port	Distance from SO
<i>Hobart</i>	Exposure					
	Population index	−0.71				
	Invasive species	−0.72	0.86			
	Distance from port	0.73	−0.91	−0.83		
	Distance from SO	−0.12	−0.36	−0.18	0.29	
	Heavy metals	−0.83	0.85	0.89	−0.81	−0.10
<i>Melbourne</i>	Exposure					
	Population index	−0.22				
	Distance from port	0.44	−0.62			
	Distance from SO	0.32	−0.38	NA	0.35	
	Heavy metals	−0.31	0.14	NA	−0.15	−0.73
<i>Sydney</i>	Exposure					
	Population index	−0.43				
	Distance from port	0.51	−0.76	NA		
	Heavy metals	−0.71	0.57	NA	−0.63	NA

Loss of native rocky reef biodiversity in Australian metropolitan embayments

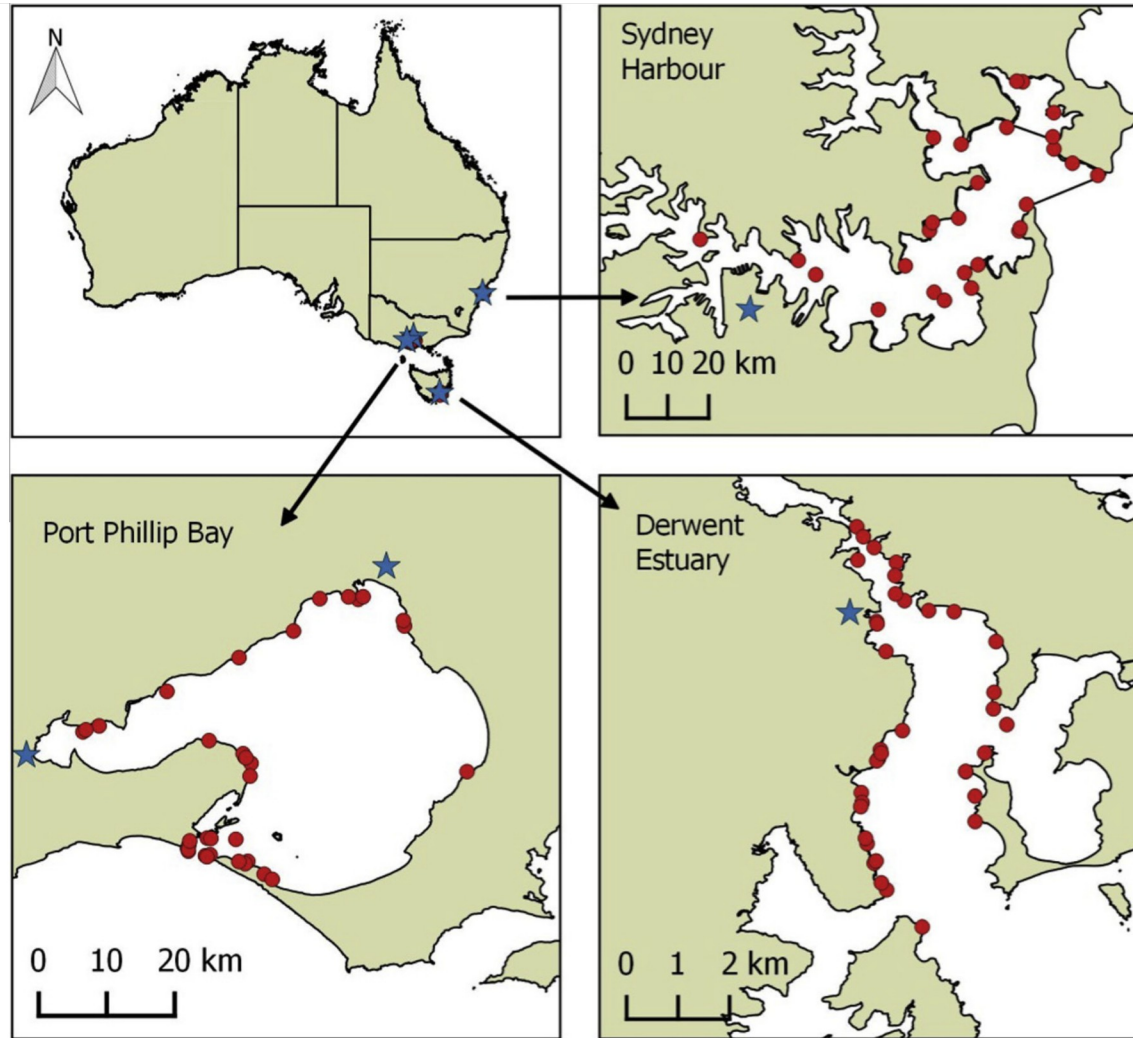


Fig. 1. Maps of study embayments and sites in south eastern Australia (circles). Stars represent locations of city ports and central business districts.

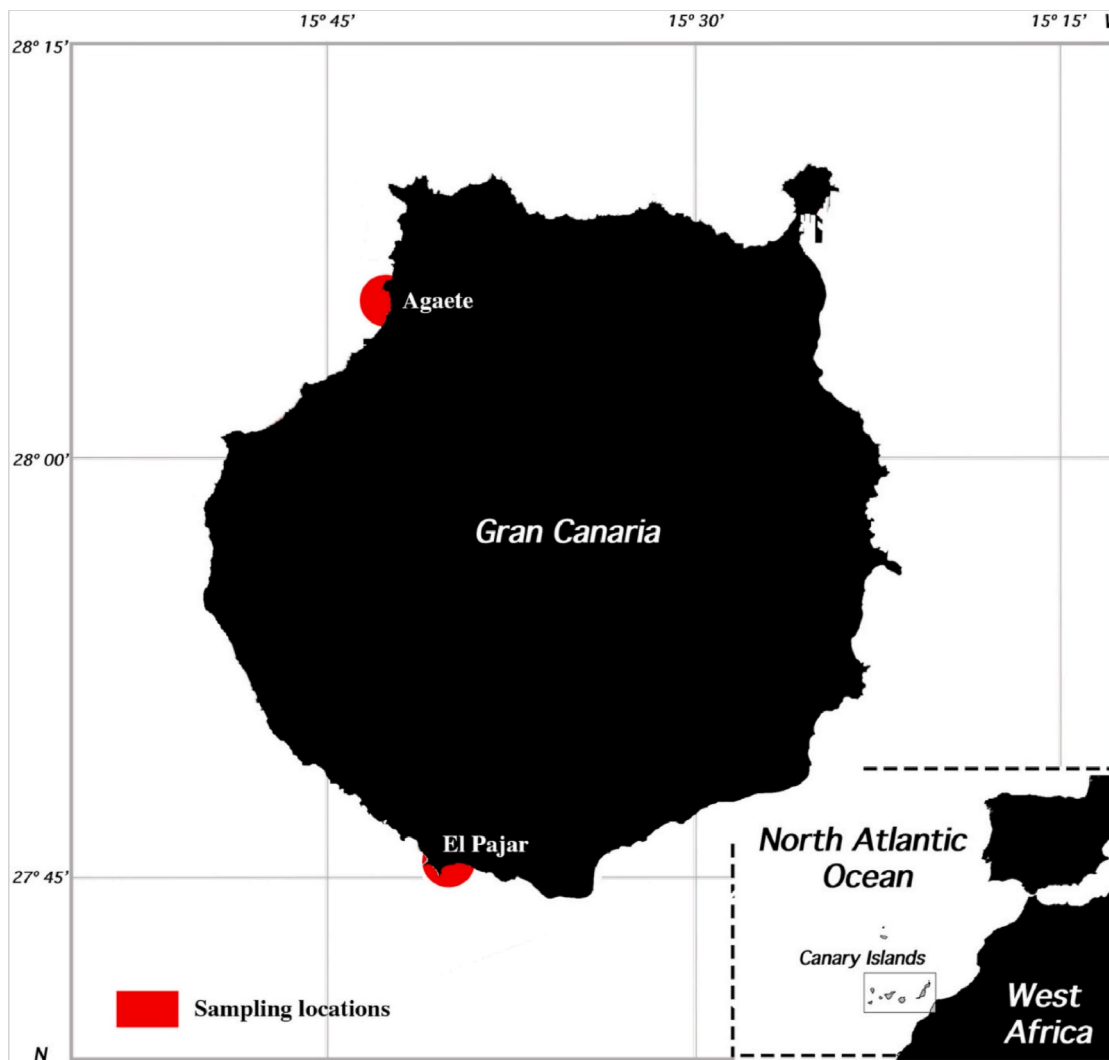


Fig. 1. Location of the two study (collecting) sites on the Gran Canaria Island coast (Canary Islands, central-east Atlantic). Playa de las Nieves (Agaete) ($28^{\circ}05'58.9''\text{N}$ $15^{\circ}42'37.4''\text{W}$) in the northern part and Santa Agueda Bay (El Pajar) in the southern part ($27^{\circ}45'02.4''\text{N}$ $15^{\circ}40'13.4''\text{W}$) of this island.

Figures and tables: more rules and conventions

1. Figures and tables are numbered consecutively, in separate series
 - Figure 1; Figure 2; Figure 3; Table 1; Table 2; Table 3
 - Can re-use numbers for a figure and table, but not two figures or two tables
2. All figures and tables must be referenced in body
 - “Grain size was large at the inflow and decreased farther out (Fig. 1).”
 - Avoid “Fig. 1 shows differences in grain size”.
3. Figures and tables are numbered in order of their appearance in the body
4. Abbreviate in-text citations (Fig. 1) and write out in caption (Figure 1). “Table” is never abbreviated.
5. Figures and tables are given at the end of a manuscript, after references. *One item per page.*
 - The journal staff arrange them if your manuscript is accepted
6. Tables should be formattable in Word- do not use a picture of a table. Paste from Excel.

Checklist for your tables and figures

1. Is every table and figure necessary?
 - Can a sentence work instead?
 - Are any figures/tables redundant with each other (showing same result)?
2. Does every figure show the correct independent/dependent variable?
3. Do you use a figure to show a pattern, trend or relationship? Do you use a table when the precision matters and you are showing many variables?
4. Is every figure/table and its caption understandable without referring to the body?
5. Are figures and tables numbered consecutively in separate series?
6. Is every figure/table referred to in the body?
7. Are figures numbered according to the order they appear in the text?
8. Does every figure/table have a caption? Does the caption appear below figures and above tables?
9. Are figures and tables given at the end of the manuscript?

More guidelines for the Results section

- **You should be able to read the text** without looking at the tables and figures **and still understand the paper.** Do not rely on figures to tell the story. Figures are enhancements!
- For each result you give in this section, there should be a corresponding set of methods in the preceding section
- **Just the facts**: save biological interpretation for the Discussion
 - Numerical interpretation good:

“The mean mercury concentration of landlocked salmon was higher than seagoing salmon (mean=, p value).” *(this is a fact)*
 - Real-world interpretation bad:

“This suggests that landlocked salmon live in more contaminated environments than seagoing salmon.”
(this is not a fact)

Even more guidelines for the Results section

- The Results section is usually short (remember- just the facts!)
- Good idea to give a one or two-sentence summary of the results, either as the first (preferably) or the last sentence
- If there are multiple experiments or types of analyses, good idea to group under subheadings that match the Methods subheadings.
- Always use the past tense!

Assignments for today and for home

In class:

1. Work on the Results section of your paper. You will submit intro, methods, results and one figure on Friday.

Outside of class:

2. Work on the Results section of your paper. You will submit intro, methods, results and one figure on Friday.
3. Watch videos on “better poster”.
4. Find a poster around FSH, FTR, MAR or elsewhere, take a photo of it, and submit it on canvas alongside three positive comments and three ideas for improvement.

Next class, we will shift gears to discuss your poster project.